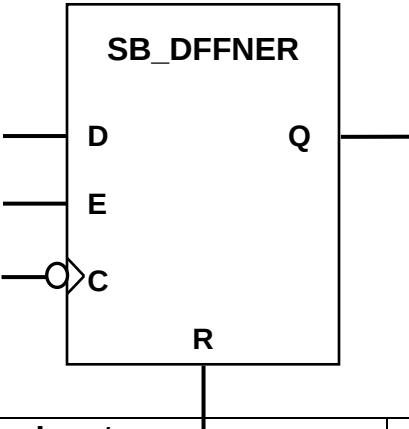


SB_DFFNER

D Flip-Flop – Negative Edge Clock, Enable and Asynchronous Reset

Data: D is loaded into the flip-flop when R is low and E is high during the falling clock edge transition.

Reset: R input is active high, and it overrides all other inputs and asynchronously resets the Q output.



Inputs				Output
R	E	D	C	Q
1	X	X	X	0
0	0	X	X	Previous Q
0	1	0	↘	0
0	1	1	↘	1
Power on State	X	X	X	0

- Key
- ↘ Falling Edge
 - 1 High logic level
 - 0 Low logic level
 - X Don't care
 - ? Unknown

HDL Usage

This register is inferred during synthesis and can also be explicitly instantiated.

Default Signal Values

The iCEcube2 software assigns the following signal values to unconnected input ports:

- Input D: Logic '0'
- Input C: Logic '0'
- Input R: Logic '0'
- Input E: Logic '1'

Note that explicitly connecting a Logic '1' value to port E will result in a non-optimal implementation, since an extra LUT will be used to generate the Logic '1'. If the user's intention is to keep the FF always enabled, it is recommended that either port E be left unconnected, or the corresponding FF without a Clock Enable port be used.

Verilog Instantiation

// SB_DFFNER - D Flip-Flop – Negative Edge Clock, Reset is asynchronously
// on falling clock edge and Clock Enable.

```
SB_DFFNER    SB_DFFNER_inst (
    .Q(Q),           // Registered Output
    .C(C),           // Clock
    .E(E),           // Clock Enable
    .D(D),           // Data
    .R(R)            // Asynchronously Reset
);
```

// End of SB_DFFNER instantiation

VHDL Instantiation

-- SB_DFFNER - D Flip-Flop – Negative Edge Clock, Reset is asynchronously
-- on falling clock edge and Clock Enable.

```
SB_DFFNER_inst:    SB_DFFNER
    port map (
        Q => Q,           -- Registered Output
        C => C,           -- Clock
        E => E,           -- Clock Enable
        D => D,           -- Data
        R => R            -- Asynchronously Reset
    );
```

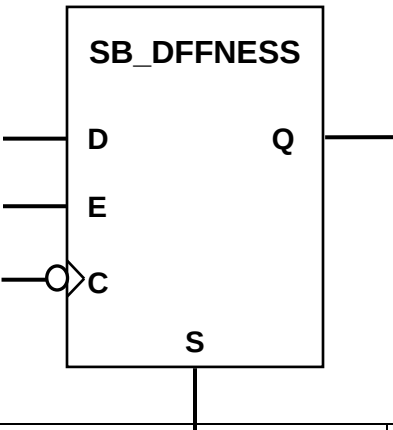
-- End of SB_DFFNER instantiation

SB_DFFNESS

D Flip-Flop – Negative Edge Clock, Enable and Synchronous Set

Data: D is loaded into the flip-flop when S is low and E is high during the falling clock edge transition.

Set: S and E inputs high, synchronously sets the Q output on the falling clock edge transition.



Inputs				Output
S	E	D	C	Q
1	1	X	↘	1
X	0	X	X	Previous Q
0	1	0	↘	0
0	1	1	↘	1
Power on State	X	X	X	0

Key
↘ Falling Edge
1 High logic level
0 Low logic level
X Don't care
? Unknown

HDL Usage

This register is inferred during synthesis and can also be explicitly instantiated.

Default Signal Values

The iCEcube2 software assigns the following signal values to unconnected input ports:

Input D: Logic '0'
Input C: Logic '0'
Input S: Logic '0'
Input E: Logic '1'

Note that explicitly connecting a Logic '1' value to port E will result in a non-optimal implementation, since an extra LUT will be used to generate the Logic '1'. If the user's intention is to keep the FF always enabled, it is recommended that either port E be left unconnected, or the corresponding FF without a Clock Enable port be used.

Verilog Instantiation

// SB_DFFNESS - D Flip-Flop – Negative Edge Clock, Set is synchronous with falling clock edge,
// and Clock Enable.

```
SB_DFFNESS    SB_DFFNESS_inst (
    .Q(Q),      // Registered Output
    .C(C),      // Clock
    .E(E),      // Clock Enable
    .D(D),      // Data
    .S(S)       // Synchronously Set
);
```

// End of SB_DFFNESS instantiation

VHDL Instantiation

-- SB_DFFNESS - D Flip-Flop – Negative Edge Clock, Set is synchronous with falling clock edge,
-- and Clock Enable.

```
SB_DFFNESS_inst : SB_DFFNESS
    port map (
        Q => Q,      -- Registered Output
        C => C,      -- Clock
        E => E,      -- Clock Enable
        D => D,      -- Data
        S => S       -- Synchronously Set
    );
```

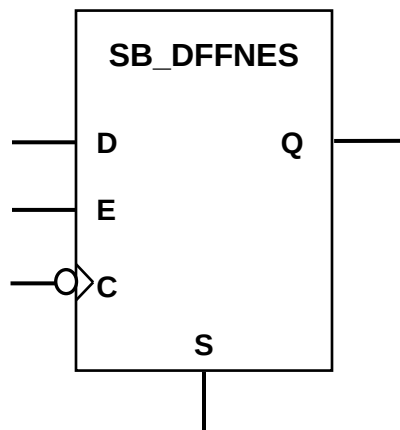
-- End of SB_DFFNESS instantiation

SB_DFFNES

D Flip-Flop – Negative Edge Clock, Enable and Asynchronous Set

Data: D is loaded into the flip-flop when S is low and E is high during the falling clock edge transition.

Set: S input is active high, and it overrides all other inputs and asynchronously sets the Q output.



Inputs				Output
S	E	D	CLK	Q
1	X	X	X	1
0	0	X	X	Previous Q
0	1	0		0
0	1	1		1
Power on State	X	X	X	0

Key
 Falling Edge
1 High logic level
0 Low logic level
X Don't care
? Unknown

HDL Usage

This register is inferred during synthesis and can also be explicitly instantiated.

Default Signal Values

The iCEcube2 software assigns the following signal values to unconnected input ports:

- Input D: Logic '0'
- Input C: Logic '0'
- Input S: Logic '0'
- Input E: Logic '1'

Note that explicitly connecting a Logic '1' value to port E will result in a non-optimal implementation, since an extra LUT will be used to generate the Logic '1'. If the user's intention is to keep the FF always enabled, it is recommended that either port E be left unconnected, or the corresponding FF without a Clock Enable port be used.

Verilog Instantiation

// SB_DFFNES - D Flip-Flop – Negative Edge Clock, Set is asynchronous on falling clock edge with clock
// Enable.

```
SB_DFFNES    SB_DFFNES_inst (
    .Q(Q),           // Registered Output
    .C(C),           // Clock
    .E(E),           // Clock Enable
    .D(D),           // Data
    .S(S)            // Asynchronously Set
);
```

// End of SB_DFFNES instantiation

VHDL Instantiation

-- SB_DFFNES - D Flip-Flop – Negative Edge Clock, Set is asynchronous
-- on falling clock edge and Clock Enable.

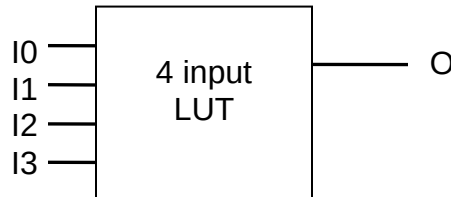
```
SB_DFFNES_inst:    SB_DFFNES
    port map (
        Q => Q,           -- Registered Output
        C => C,           -- Clock
        E => E,           -- Clock Enable
        D => D,           -- Data
        S => S            -- Asynchronously Set
    );
```

-- End of SB_DFFNES instantiation

Combinational Logic Primitives

SB_LUT4

The LUT unit is a simple ROM 4 input look-up function table.



Initialization values

LUT state initialization parameter LUT_INIT = 16'hxxxx;

Inputs				Output
I3	I2	I1	I0	O
0	0	0	0	LUT_INIT[0]
0	0	0	1	LUT_INIT[1]
0	0	1	0	LUT_INIT[2]
0	0	1	1	LUT_INIT[3]
0	1	0	0	LUT_INIT[4]
0	1	0	1	LUT_INIT[5]
0	1	1	0	LUT_INIT[6]
0	1	1	1	LUT_INIT[7]
1	0	0	0	LUT_INIT[8]
1	0	0	1	LUT_INIT[9]
1	0	1	0	LUT_INIT[10]
1	0	1	1	LUT_INIT[11]
1	1	0	0	LUT_INIT[12]
1	1	0	1	LUT_INIT[13]
1	1	1	0	LUT_INIT[14]
1	1	1	1	LUT_INIT[15]

HDL Usage

This primitive is inferred during synthesis and can also be explicitly instantiated.

Default Signal Values

The iCEcube2 software assigns logic value '0' to unconnected input ports.

Verilog Instantiation

```
// SB_LUT4 : 4-input Look-Up Table

SB_LUT4      SB_LUT4_inst (
    .O (O),           // output
    .I0 (I0),         // data input 0
    .I1 (I1),         // data input 1
    .I2 (I2),         // data input 2
    .I3 (I3)          // data input 3
);
defparam SB_LUT4_inst.LUT_INIT=16'hxxxx;
                //LUT state initialization parameter, 16 bits.

//End of SB_LUT4 instantiation
```

VHDL Instantiation

```
-- SB_LUT4 : 4-input Look-Up Table

SB_LUT4_inst: SB_LUT4
    generic map(
        LUT_INIT => x"0001"    -- LUT state initialization parameter, 16 bits
    )
    port map (
        I0 => I0,
        I1 => I1,
        I2 => I2,
        I3 => I3,
        O  => O
    );
```

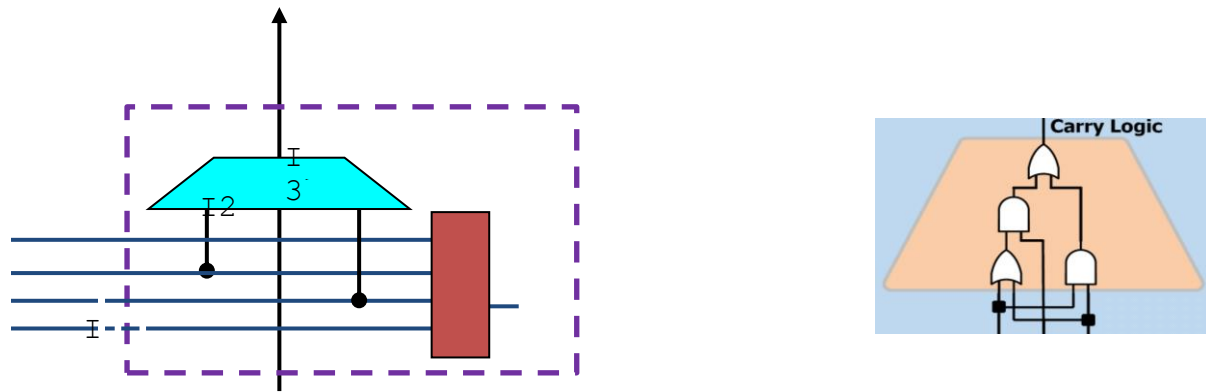

SB_CARRY

Carry Logic

The dedicated Carry Logic within each Logic Cell primarily accelerates and improves the efficiency of arithmetic logic such as adders, accumulators, subtractors, incrementers, decrementers, counters, ALUs, and comparators. The Carry Logic also supports a limited number of wide combinational logic functions.

The figure below illustrates the Carry Logic structure within a Logic Cell. The Carry Logic shares inputs with the associated Look-Up Table (LUT). The I1 and I2 inputs of the LUT directly feed the Carry Logic.. The carry input from the previous adjacent Logic Cell optionally provides an alternate input to the LUT4 function, supplanting the I3 input.

Carry Logic Structure within a Logic Cell



Inputs			Output
I0	I1	CI	CO
0	0	X	0
0	X	0	0
X	1	1	1
X	0	0	0
1	X	1	1
1	1	X	1

HDL Usage

This primitive is inferred during synthesis and can also be explicitly instantiated.

Default Signal Values

The iCEcube2 software assigns logic value '0' to unconnected input ports.

Verilog Instantiation

```
SB_CARRY my_carry_inst (  
    .CO(CO),  
    .I0(I0),  
    .I1(I1),  
    .CI(CI));
```

VHDL Instantiation

```
my_carry_inst : SB_CARRY  
    port map (  
        CO => CO,  
        CI => CI,  
        I0 => I0,  
        I1 => I1  
    );
```